Fourier Transform and PDE's (Chapter 16)

We have spent a great deal of time studying the 1-d heat equation with various boundary conditions. However, in all cases we used a rod which had finite length. Being good math students we might ask, what about an infinite rod?

Let's Examine the heat equation defined on the interval $(-\infty, \infty)$

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

Let's proceed in the usual way by separating variables. With our usual assumption $u(x,t) = h(t)\phi(x)$ we obtain our old friends:

$$\frac{dh}{dt} = -\lambda k h \qquad\qquad \frac{d^2\phi}{dx^2} = -\lambda\phi$$

With our usual solutions:

$$h(t) = c_1 e^{-\lambda k t} \qquad\qquad \phi(x) = c_1\cos\left(\sqrt{\lambda}x\right) + c_2\sin\left(\sqrt{\lambda}x\right)$$

Normally at this point we would apply our boundary conditions to determine our separation constant λ alas we have no boundaries as this is an infinite interval. So instead we will make the requirement that:

$$u(\pm\infty, t) < \infty \quad \Rightarrow \quad |\phi(\pm\infty)| < \infty$$

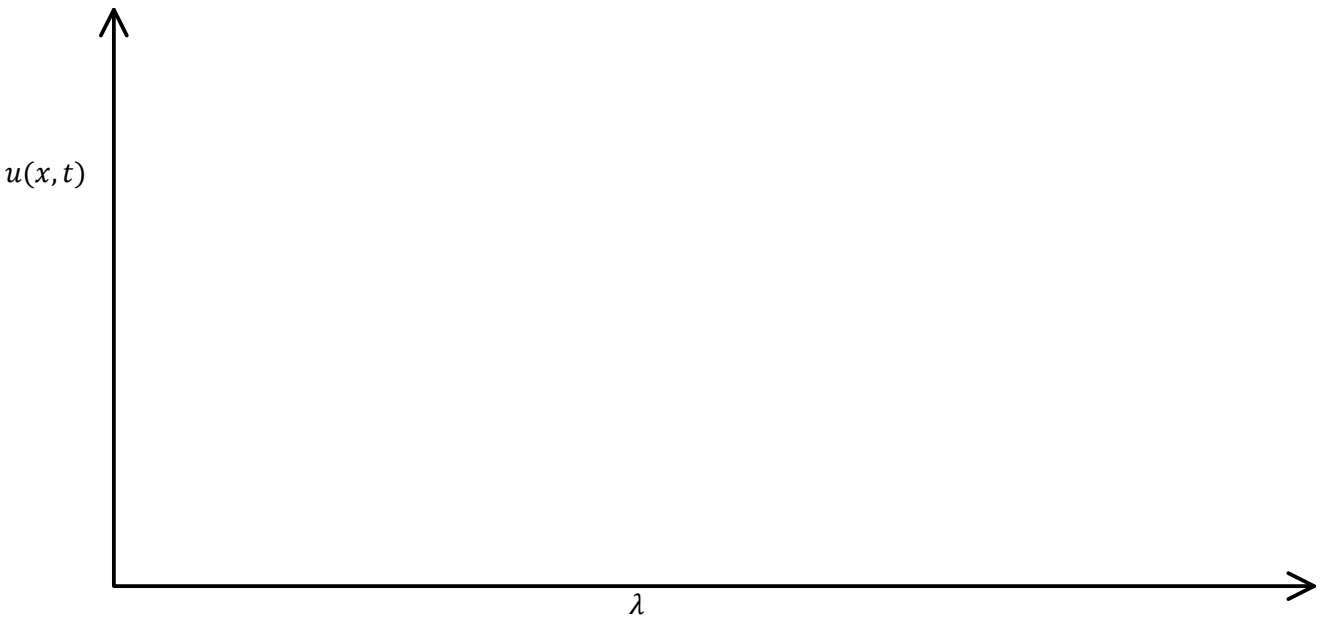
The question then becomes, for what values of λ do we satisify $|\phi(\pm\infty)| < \infty$

So as long as $\lambda \geq 0$ our condition of $|\phi(\pm\infty)| < \infty$ is satisfied, this is very different from the case we had before where $\lambda = \left(\frac{n\pi}{L}\right)^2$. We often refer to the set of eigen values as the spectra of a problem. When we had a finite rod we had a discrete spectra for this problem we have a continuious spectra.

Discrete Spectra $\lambda = \left(\frac{n\pi}{L}\right)^2$

Continuous Spectra $\lambda \geq 0$

So now we have a solution $u(x,t) = h(t)\phi(x) = e^{-k\lambda t}\left(c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right)\right)$ for each value of $\lambda \geq 0$, so again we have infinately many solutions but with a continuious spectra, the question now is how do we apply the principle of super position.



With these considerations instead of summing over λ we integrate over λ

$$u(x,t) = \int_0^\infty e^{-k\lambda t}\left(c_1(\lambda)\cos\left(\sqrt{\lambda}x\right) + c_2(\lambda)\sin\left(\sqrt{\lambda}x\right)\right) d\lambda$$

Possible Final Problem:

Verify that the solution $u(x,t) = \int_0^\infty e^{-k\lambda t}\left(c_1(\lambda)\cos\left(\sqrt{\lambda}x\right) + c_2(\lambda)\sin\left(\sqrt{\lambda}x\right)\right) d\lambda$ satisfies the 1-d heat equation:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

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So now that we have our solution we need to find the coefficients. We assume we have some initial condition $u(x,0) = f(x)$ and proceed in the usual way.

$$u(x,t) = \int_0^\infty e^{-k\lambda t}\left(c_1(\lambda)\cos\left(\sqrt{\lambda}x\right) + c_2(\lambda)\sin\left(\sqrt{\lambda}x\right)\right) d\lambda$$

Real quickly, since lambda is positive anyway we will define $\lambda = \omega^2$ and write our solution in the form.

$$u(x,t) = \int_0^\infty e^{-k\omega^2 t}(A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x))\, d\omega$$

Let's now apply our initial condition:

$$u(x,0) = f(x) = \int_0^\infty (A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x))\, d\omega$$

The question then becomes how do we find $A(\omega)$ and $B(\omega)$, this is where the Fourier transform will come in.

Before we do that though we need to re-write our initial condition equation and solution using complex exponentials

It can be shown that:

$$u(x,t) = \int_0^\infty e^{-k\omega^2 t}(A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x))\, d\omega = \int_{-\infty}^\infty c(\omega)e^{-i\omega x}e^{-k\omega^2 t}d\omega$$

$$f(x) = \int_0^\infty (A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x))\, d\lambda = \int_{-\infty}^\infty c(\omega)e^{-i\omega x}d\omega$$

with $A(\omega), B(\omega) \in R$ and $c(\omega) \in C$

In fact a great possible final question would be to determine $c(\omega)$ in terms of $A(\omega)$ and $B(\omega)$ using the equation for $u(x,t)$ above!

For now let's just use the fact that we could write:

$$u(x,t) = \int_{-\infty}^\infty c(\omega)e^{-i\omega x}e^{-k\omega^2 t}d\omega \qquad f(x) = \int_{-\infty}^\infty c(\omega)e^{-i\omega x}d\omega$$

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In order to find the coefficient $c(\omega)$ we now introduce the Fourier Transform and the Inverse Fourier Transform

The Fourier transformation of a function $f(x)$ is defined as:

$$\mathcal{F}[f(x)] = F(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{i\omega x}dx$$

The inverse Fourier transform of a function $F(\omega)$ is defined as:

$$\mathcal{F}^{-1}[F(\omega)] = f(x) = \int_{-\infty}^{\infty} F(\omega)e^{-i\omega x}d\omega$$

Let's examine our initial condition equation:

$$f(x) = \int_{-\infty}^{\infty} c(\omega)e^{-i\omega x}d\omega$$

Looking at this we notice that this in the inverse Fourier Transform of $c(\omega)$! What does this say about $c(\omega)$?

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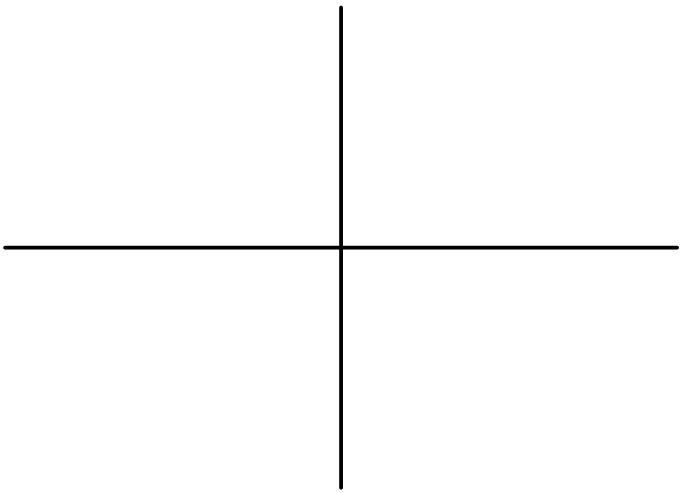
So we have a solution to our infinite domain heat equation:

$$u(x,t) = \int_{-\infty}^{\infty} c(\omega) e^{-i\omega x} e^{-k\omega^2 t} d\omega$$ where: $$c(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x) e^{i\omega x} dx$$

It might be useful to see an example of the calculation of a Fourier transform. Suppose $f(x) = e^{-\alpha x^2}$ is our initial condition.

Let's find our $c(\omega)$

$$c(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\alpha x^2} e^{i\omega x} dx$$



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It turns out we use Fourier Transforms to turn PDE's into ODE's! Let's do this with our heat equation.

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

The trick will be to take the Fourier Transform of both sides of this equation. Before we do that let's just name the Fourier Transform of $u(x,t)$ to be $\overline{U}(\omega,t)$ that is:

$$\mathcal{F}[u(x,t)] = \overline{U}(\omega,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} u(x,t)e^{i\omega x}dx$$

Now Let's see what happens when we take the Fourier Transform of both sides of our PDE.

The time derivative side is easy

For the Spatial derivative side Let's first examine what happens when we take the Fourier Transform of $\frac{\partial u}{\partial x}$

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$$\mathcal{F}\left[\frac{\partial u}{\partial x}\right] = \int_{-\infty}^{\infty} \frac{\partial u}{\partial x} e^{i\omega x} dx$$

Let's throw integration by parts at it.

So $\mathcal{F}\left[\frac{\partial u}{\partial x}\right] = -i\omega\mathcal{F}[u]$

Any guesses for $\mathcal{F}\left[\frac{\partial^2 u}{\partial x^2}\right]$?

Then if we take the Fourier transform of our heat equation on both sides we get:

$$\mathcal{F}\left[\frac{\partial u}{\partial t}\right] = \mathcal{F}\left[k\frac{\partial^2 u}{\partial x^2}\right]$$

$$\frac{\partial}{\partial t}\overline{U}(\omega, t) = k(-i\omega)^2\overline{U}(\omega, t)$$

An ODE! Let's solve this thing.

So we have $\overline{U}(\omega, t) = c(\omega)e^{-k\omega^2 t}$ we can then simply apply the inverse Fourier Transform to obtain $u(x, t)$ That is

$$u(x, t) = \int_{-\infty}^{\infty} \overline{U}(\omega, t)e^{-i\omega t}d\omega = \int_{-\infty}^{\infty} c(\omega)e^{-k\omega^2 t}\, e^{-i\omega t}d\omega$$

Just as we had from separation of variables!

We could find $c(\omega)$ again using:

$$c(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{i\omega x}dx$$

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Another representation for our solution using the Convolution Theorem. First Let's define the convolution of two functions, $f * g$

$$f * g = \int_{-\infty}^{\infty} f(\bar{x})g(x - \bar{x})d\bar{x}$$

Let's ask the following question, What is the inverse Fourier transform h(x) of a function $H(\omega) = F(\omega)G(\omega)$

Where: $F(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{-i\omega x}dx$ $G(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-i\omega x}dx$

$f(x) = \int_{-\infty}^{\infty} F(\omega)e^{i\omega x}d\omega$ $g(x) = \int_{-\infty}^{\infty} G(\omega)e^{i\omega x}d\omega$

We start by just applying the inverse Fourier transform to $H(\omega)$ $h(x) = \int_{-\infty}^{\infty} H(\omega)e^{i\omega x}d\omega = \int_{-\infty}^{\infty} F(\omega)G(\omega)e^{i\omega x}d\omega$

So we have that if $H(\omega) = F(\omega)G(\omega)$ then $h(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(\bar{x})f(x-\bar{x})d\bar{x} = g * f$

In words we are just saying that inverse Fourier transform of a product of two Fourier transforms is $\frac{1}{2\pi}$ times the convolution of of the product of the orignal functions.

What does this do for us with regard to the heat equation? Well recall that the solution was:

$$u(x,t) = \int_{-\infty}^{\infty} c(\omega)e^{-k\omega^2 t}\, e^{-i\omega t}d\omega = \mathcal{F}^{-1}[c(\omega)\, e^{-k\omega^2 t}]$$

Let's name $G(\omega) = c(\omega)$ and $F(\omega) = e^{-k\omega^2 t}$

Then by the convolution theorem $u(x,t)$ will be equal to the $\frac{1}{2\pi}$ times convolution of the inverse Fourier transforms of $G(\omega) = c(\omega)$ and $F(\omega) = e^{-k\omega^2 t}$

We know that the inverse Fourier transform of $c(\omega)$ is just of $f(x)$ and we found out earlier that:

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\alpha x^2}e^{i\omega x}dx = \frac{1}{\sqrt{4\pi\alpha}}e^{-\frac{\omega^2}{4\alpha}}$$

If we take the Inverse Fourier transform of $F(\omega) = e^{-k\omega^2 t}$ it would read: $\int_{-\infty}^{\infty} e^{-k\omega^2 t}\, e^{-i\omega x}d\omega$ which we can figure out easily using the known integral above. Let $\alpha = kt, x = -\omega$

so we have that $\mathcal{F}^{-1}[c(\omega)] = f(x)$ and $\mathcal{F}^{-1}\left[e^{-k\omega^2 t}\right] = \sqrt{\frac{\pi}{kt}}e^{-\frac{x^2}{4kt}}$

What we want is $\mathcal{F}^{-1}\left[c(\omega)e^{-k\omega^2 t}\right]$

The convolution Theorem States that:

$$\mathcal{F}^{-1}[H(\omega)] = \mathcal{F}^{-1}[F(\omega)G(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(\bar{x})f(x-\bar{x})d\bar{x}$$

where $g(x) = \mathcal{F}^{-1}[F(\omega)] \quad and \quad f(x) = \mathcal{F}^{-1}[G(\omega)]$

If we apply this to our solution

$$\bar{U}(\omega, t) = c(\omega)e^{-k\omega^2 t}$$